



FACULTY OF INFORMATION TECHNOLOGY
BACHELOR OF SCIENCE IN INFORMATICS AND COMPUTER SCIENCE
END OF SEMESTER EXAMINATION
ICS 1202: OBJECT-ORIENTED PROGRAMMING I

Date: 19th March 2018

Time: 2 Hours

Instructions

1. This examination consists of **FIVE** questions.
2. Answer **Question ONE (COMPULSORY)** and any other **TWO** questions.

Question One [30 Marks]

- a) Explain the following terms as used in object-oriented programming: **(10 Marks)**
 - i. Attribute
 - ii. Behavior
 - iii. Event
 - iv. Listener
 - v. API
- b) Class A has attributes x and y. Class B has attributes m and n, and it also inherits class A. List the attributes of the objects shown below: **(4 Marks)**
 - i. B obj1 = new A();
 - ii. A obj2 = new B();
- c) Explain the term event-handling as used in Java programming. Using a diagram, illustrate how event-handling can be implemented in a Java program. **(6 Marks)**
- d) Using Java programming code segments, illustrate how a class can implement an interface. **(10 Marks)**

Question Two [15 Marks]

- a) Explain the difference between encapsulation and data abstraction as used in object-oriented programming. **(4 Marks)**
- b) Indicate true/false in the following statements. Explain your answer. **(6 Marks)**
- Classes and objects mean the same thing
 - Both classes and objects have attributes and behaviors
 - Data hiding in Java programming means hiding Java files
- c) Write the Java code for class Circle as shown in the UML. **(5 Marks)**

Cylinder extends Circle
-height: int
+Cylinder(radius: int, height: int) +getArea(): float +getPerimeter(): float

Question Three [15 Marks]

- a) Describe the term UML as used in class designs. **(2 Marks)**
- b) Using a diagram, illustrate the three main parts of a UML. **(6 Marks)**
- c) Write a simple Java application that will prompt the user to enter two numbers, the program should display even numbers between the two numbers. **(7 Marks)**

Question Four [15 Marks]

- a) Explain the significance of exception-handling as used in Object-oriented programming. **(2 Marks)**
- b) Explain the difference between throw and throws as used in Object-oriented programming. **(4 Marks)**
- c) Identify 3 errors in the code segments below. **(6 Marks)**

```
public class Error {  
  
    public static void main(String args[]){  
        try{  
            Exception obj;  
            obj = new Exception("I am an error");  
            throw obj;  
        }catch(Error e){  
            System.println(e.getMessage());  
        }final{  
            System.out.println("No errors found");  
        }  
    }  
}
```

- d) Write the correct Java codes for class Error. **(3 Marks)**

Question Five [15 Marks]

- a) Define the term relational database. **(2 Marks)**
- b) Explain the main difference between SQL and Java Language. **(4 Marks)**
- c) List down at least 3 Java classes that are used to connect a Java application to a database. **(3 Marks)**
- d) Using a diagram, discuss the JDBC architecture. **(6 Marks)**